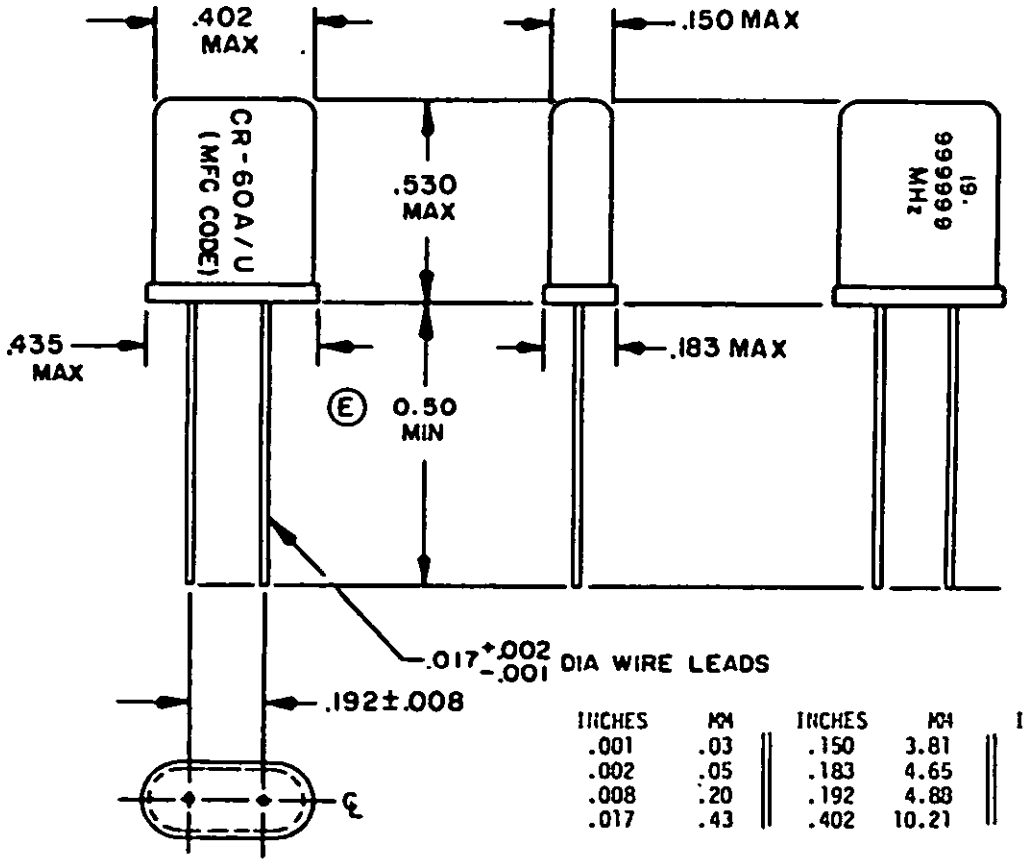
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-60A/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 5 to 20 MHz; fundamental; noncontrolled; series.



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.150	3.81	.435	11.05
.002	.05	.183	4.65	.50	12.7
.008	.20	.192	4.88	.530	13.46
.017	.43	.402	10.21		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The type-designation, frequency, and code-designating characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Dimensions and configuration.

E denotes changes

FSC 5955

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TABLE I. Requirements.

Requirements	Specified value or condition
Holder	HC-18/U
Dimensions, marking, and configuration	See figure
Frequency range	5 to 20 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Frequency tolerance:	
Operating temperature range	±50 PPM
Equivalent resistance	See table III
Mode of oscillation	Fundamental
Test set, calibration values, and rated drive level	See table II
Operating temperature range	-55°C to +105°C, inclusive
Shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 PPM

1/ Method 204 of MIL-STD-202, test condition A.

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mW
TS-330/TSM	5 to 7.5	25	14	---	5.0 ±1.0
	7.5+ to 10	16	18	---	5.0 ±1.0
	10+ to 15	13	20	---	5.0 ±1.0
TS-683/TSM	15+ to 20	12	---	0.24	5.0 ±1.0

TABLE III. Equivalent resistance.

Frequency range, inclusive	Maximum resistance
MHz	Ohms
5 to 7	50
7+ to 10	30
10+ to 15	25
15+ to 20	20

NOTE: NATO preferred type NXT21.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

Users activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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